```
# ========================================
#       BAKERY BASIC STOCK INVENTORY SYSTEM
# ========================================

# AUTHORIZED USERS
AUTHORIZED_USERS = {
    "owner": {"password": "owner123", "role": "Owner"},
    "staff": {"password": "staff123", "role": "Staff"}
}

# MASTER DATA STORAGE
inventory = []

# VALIDATION LISTS
ALLOWED_PRODUCTS = [
    "flour", "all-purpose flour", "bread flour", "cake flour",
    "sugar", "white sugar", "brown sugar", "powdered sugar",
    "milk", "fresh milk", "evaporated milk", "condensed milk",
    "eggs", "fresh eggs", "butter", "margarine",
    "yeast", "baking powder", "baking soda",
    "vanilla", "cocoa powder", "chocolate",
    "oil", "vegetable oil", "packaging", "boxes", "bags", "labels"
]

# CATEGORY-SPECIFIC BRANDS
BRANDS_FLOUR = [
    "maya", "philmills", "golden prize",
    "king arthur", "baker's choice"
]

BRANDS_SUGAR = [
    "central azucarera", "united sugar",
    "davao sugar", "cagayan sugar", "fil-sugar"
```

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]

BRANDS_CONDENSED_MILK = [
    "nestle", "carnation", "alaska",
    "magnolia", "dutch lady"
]

BRANDS_BUTTER = [
    "magnolia", "alaska", "anchor",
    "emborg", "golden churn"
]

ALLOWED_BRANDS = (
    BRANDS_FLOUR + BRANDS_SUGAR +
    BRANDS_CONDENSED_MILK + BRANDS_BUTTER
)

ALLOWED_CATEGORIES = [
    "flour", "sugar", "dairy", "eggs",
    "fats", "leavening", "flavoring",
    "packaging", "other"
]

ALLOWED_UNITS = [
    "kg", "g", "l", "ml", "pcs", "pack", "box"
]



# ========================================
#             HELPER FUNCTIONS
# ========================================

def press_enter_to_continue():
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    input("\nPress Enter to continue...")



def confirm_proceed():
    while True:
        ans = input("Proceed? (1=Yes | 2=No): ").strip()

        if ans == "1":
            return True
        elif ans == "2":
            return False
        else:
            print("Invalid! Enter only 1 or 2.")



def get_valid_number(prompt):
    while True:
        try:
            num = float(input(prompt).strip())

            if num > 0:
                return num

            print("Value must be greater than 0.")

        except ValueError:
            print("Enter a valid number only.")



def display_options_and_select(prompt, options):
    while True:
        print()
```

```
        for idx, item in enumerate(options, 1):
            print(f"{idx}. {item.title()}")

        choice = input(f"\n{prompt}: ").strip()

        if choice.isdigit():
            pos = int(choice) - 1

            if 0 <= pos < len(options):
                return options[pos]

        print("Invalid selection! Choose from the list.")


def get_valid_product():
    return display_options_and_select(
        "Enter product number", ALLOWED_PRODUCTS
    )


def get_brand_for_product(product_name):
    product = product_name.lower()

    if "flour" in product:
        brands = BRANDS_FLOUR

    elif "sugar" in product:
        brands = BRANDS_SUGAR

    elif "condensed milk" in product:
        brands = BRANDS_CONDENSED_MILK

    elif "butter" in product:
```

```
        brands = BRANDS_BUTTER

    elif "milk" in product:
        brands = BRANDS_CONDENSED_MILK

    else:
        brands = ALLOWED_BRANDS

    return display_options_and_select(
        "Select brand number", brands
    )


def get_valid_category():
    return display_options_and_select(
        "Enter category number", ALLOWED_CATEGORIES
    )


def get_valid_unit():
    return display_options_and_select(
        "Enter unit number", ALLOWED_UNITS
    )


# ========================================
#               LOGIN SYSTEM
# ========================================

def login():
    print("\n" + "=" * 50)
    print("       BAKERY INVENTORY SYSTEM - LOGIN")
    print("=" * 50)
```

```
print("Authorized Personnel Only\n")


attempts_left = 3


while attempts_left > 0:


    print(f"Attempts remaining: {attempts_left}")


    username = input("Username: ").strip().lower()
    password = input("Password: ").strip()


    if (
        username in AUTHORIZED_USERS
        and AUTHORIZED_USERS[username]["password"] == password
    ):


        role = AUTHORIZED_USERS[username]["role"]


        print(f"\nWelcome, {role}!")
        print("Verifying access...")


        if confirm_proceed():
            print("Access Granted - System Ready!")
            return role


        print("Login cancelled.")
        return None


    else:
        attempts_left -= 1
        print("Incorrect username or password!\n")


print("Maximum login attempts reached.")
```

```
    return None



# ========================================

#              ADD INVENTORY

# ========================================


def add_item():


    while True:


        print("\n========== ADD INVENTORY ==========")


        product = get_valid_product()

        brand = get_brand_for_product(product)

        category = get_valid_category()

        quantity = get_valid_number("Enter quantity: ")

        unit = get_valid_unit()


        item = {

            "product": product,

            "brand": brand,

            "category": category,

            "quantity": quantity,

            "unit": unit

        }


        inventory.append(item)


        print("\nItem successfully added!")


        if not confirm_proceed():

            break
```

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# ========================================

#             VIEW INVENTORY

# ========================================


def view_inventory():


    print("\n========== BAKERY INVENTORY ==========")


    if not inventory:

        print("Inventory is empty!")

        return


    for index, item in enumerate(inventory, 1):


        print(f"\nItem Number: {index}")

        print(f"Product: {item['product'].title()}")

        print(f"Brand: {item['brand'].title()}")

        print(f"Category: {item['category'].title()}")

        print(f"Quantity: {item['quantity']:g}")

        print(f"Unit: {item['unit']}")




# ========================================

#             SEARCH INVENTORY

# ========================================


def search_item():


    print("\n========== SEARCH INVENTORY ==========")


    keyword = input("Enter product name: ").strip().lower()
```

```
    found = False

    for item in inventory:

        if (
            keyword in item["product"].lower()
            or keyword in item["brand"].lower()
        ):

            print("\nItem Found!")
            print(f"Product: {item['product']}")
            print(f"Brand: {item['brand']}")
            print(f"Category: {item['category']}")
            print(f"Quantity: {item['quantity']:g} {item['unit']}")

            found = True

    if not found:
        print("No matching item found.")



# ========================================
#             UPDATE INVENTORY
# ========================================

def update_item():

    view_inventory()

    if not inventory:
        return
```

```
try:
    number = int(input("\nEnter item number to update: "))

    if 1 <= number <= len(inventory):

        item = inventory[number - 1]

        print("\n1. Update Quantity")
        print("2. Update Category")
        print("3. Update Unit")

        choice = input("Choose option: ").strip()

        if choice == "1":
            item["quantity"] = get_valid_number(
                "Enter new quantity: "
            )

        elif choice == "2":
            item["category"] = get_valid_category()

        elif choice == "3":
            item["unit"] = get_valid_unit()

        else:
            print("Invalid selection.")
            return

        print("Inventory successfully updated!")

    else:
        print("Invalid item number.")
```

```
    except ValueError:
        print("Enter a valid number.")




# ========================================
#             DELETE INVENTORY
# ========================================


def delete_item():

    view_inventory()

    if not inventory:
        return

    try:
        number = int(input("\nEnter item number to delete: "))

        if 1 <= number <= len(inventory):

            if confirm_proceed():

                inventory.pop(number - 1)
                print("Item successfully deleted!")

            else:
                print("Deletion cancelled.")

        else:
            print("Invalid item number.")

    except ValueError:
        print("Enter a valid number.")
```

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# ========================================

#               MAIN MENU

# ========================================


def main_menu(role):


    while True:


        print("\n" + "=" * 45)

        print("       BAKERY BASIC STOCK INVENTORY")

        print("=" * 45)


        print(f"Logged in as: {role}")


        print("\n1. Add Inventory")

        print("2. View Inventory")

        print("3. Search Inventory")

        print("4. Update Inventory")


        if role == "Owner":

            print("5. Delete Inventory")


        print("6. Logout")


        choice = input("\nEnter your choice: ").strip()


        if choice == "1":

            add_item()


        elif choice == "2":

            view_inventory()
```

```
        elif choice == "3":
            search_item()

        elif choice == "4":
            update_item()

        elif choice == "5" and role == "Owner":
            delete_item()

        elif choice == "6":
            print("\nLogging out...")
            break

        else:
            print("Invalid choice! Please try again.")


# ==========================================
#             RUN THE SYSTEM
# ==========================================

def main():

    role = login()

    if role:
        main_menu(role)

    print("\nThank you for using Bakery Inventory System!")


main()
```